AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M251-M257
Object Oriented Programming Using Java

Final Examination
Spring Semester 2025/2026

Date: 2 / Jun. / 2026

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Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed: 3 hours**

Instructions:

1. This exam consists of three parts. It is required to solve **ALL** of them in the external answer booklet:
   - **Part 1: 5** MCQ (10 marks)
   - **Part 2: 3** short answer questions (35 marks)
   - **Part 3: 2** problem solving questions (55 marks)
2. Go through all questions before you start answering them.
3. It is not allowed to use any external material.
4. It is not allowed to use any electronic device (calculators, mobiles, …)
5. At the end of the examination:
   - Check that you have written **your name, student ID and section number** on the first page of your answer booklet
   - Place your question paper in the answer booklet and hand over them to the invigilator.

PART 1: MCQ [10 Marks]: Each question carries a weight of 2 marks

Question 1: [10 marks] Read the following questions and choose the best answer.

1. Classes declared as _____ cannot be inherited.
   a. abstract b. final c. public
   d. All the above e. None of the above

2. Which of the following has a constructor?
   a. concrete class b. abstract class c. interface
   d. A and B e. B and C

3. Which of the following is a checked Exception?
   a. FileNotFoundException b. ArithmeticException c. NullPointerException
   d. All the above e. None of the above

4. _____ places the nodes in the top, right, bottom, left and center regions.
   a. VBox b. HBox c. BorderPane
   d. GridPane e. FlowPane

5. _____ method moves the thread from the blocked state to the runnable state.
   a. sleep b. run c. yield
   d. start e. notify

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PART 2: Short Answer Questions [35 Marks]

Question 2: [12 marks: 2+2+2+2+4]

Given the following code, answer the given questions:

```
public interface Switchable {
  public void turnOn();
  public void turnOff();
}
```

```
public class Fan implements Switchable
{
  public void turnOn() {
    System.out.println("Running");
  }
  public void turnOff() {
    System.out.println("Stopped");
  }
  public void showSpeed() {
    System.out.println("High");
  }
}
```

```
public class Test {
 public static void main(String[] args)
 {
  Switchable s = new Fan();
  Fan f = new Fan();

  // valid or not?
  //1// s.showSpeed();
  //2// Switchable x = new Switchable();

  f.showSpeed();
  f.turnOn();
  s.turnOff();
  s.turnOn();
 }
}
```

A. Draw a simple class diagram representing the relationship between `Switchable` and `Fan`.
B. Extract from the code an abstract method **and** mention the name of its class (or interface).
C. **Is** the code in the line that starts with the comment `//1//` **valid**? **Explain**.
D. **Is** the code in the line that starts with the comment `//2//` **valid**? **Explain**.
E. Provide the exact output of the class `Test`.

Question 3: [13 marks]

Write the code of a method that takes a TreeSet of integers (represent the marks of students of a certain course) as an argument. The method should do the following:

- Declare and create an empty ArrayList of integers.
- Iterate over all marks in the Set and add only the Passed students (whose marks >= 50) to the ArrayList.
- Print the marks in the ArrayList in reverse order.

For example, if the Set = [30, 50, 60, 80, 90] the method should create the ArrayList [50, 60, 80, 90] and print on the screen **90 80 60 50**.

Question 4: [10 marks]

What does the following method return for **n = 6**? (Show your working steps in detail)

```
public static int mystery(int n) {
    if (n == 0) return 0;
    return 3 + mystery(n / 2);
}
```

PART 3: Problem Solving Questions [55 Marks]

Question 5: [42 marks]

Introduction:
There are many libraries in the university. Each library has many books and stores all information about the books and their quantities in a Map.

1. Write the class **Book** which includes: **[15 marks: 2+3+2+2+4+2]**
   a) 3 attributes: book ISBN (e.g. "978-0138362171"), title (e.g. "Java Programming") and number of pages (e.g. 651).
   b) a zero-arg constructor and a multi-arg constructor.
   c) getter and setter for the attribute **title**.
   d) **toString()** method which overrides the Object's toString() method.
   e) **equals()** method which overrides the Object's equals() method. It is enough to compare the book ISBN.
   f) **compareTo()** method which implements the Comparable's compareTo() method. You should use the book ISBN only in the comparison.

2. Write the class **Library** which includes: (add the necessary **import** statements) **[22 marks: 1+2+6+3+3+7]**
   a) 2 attributes: Library name and a **Map** called **books** to store **Book** objects as keys and their quantities as values. The map should be sorted according to the keys.
   b) one-arg constructor that accepts the Library name as an argument and creates a new empty Map to store the books of the library.
   c) method **addBook()** that accepts 2 arguments: a **Book** object and its quantity. It adds this pair to the map **books** and displays a message that the book has been successfully added (or an error message if the book has been added before).
   d) method **borrowBook()** that accepts a **Book** object as an argument and decreases its quantity by 1.
   e) method **returnBook()** that accepts a **Book** object as an argument and increases its quantity by 1.
   f) method **saveIntoFile()** which accepts a file name as an argument. It stores the **Library name** in a separate line then stores each book in a different line as follows:
   book title: its quantity

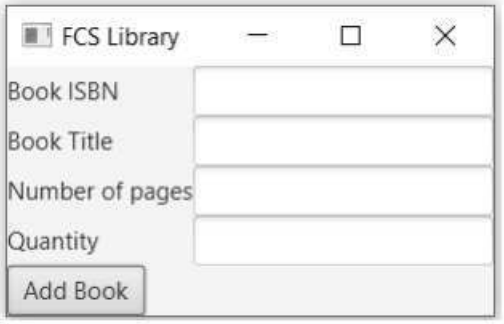
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3. Write a code to test your classes by performing the following tasks: **[5 marks]**
   a) create a **Library** object called **myLibrary** where the Library name is "**FCS Library**".
   b) create 2 **Book**s and add them with a suitable quantity to **myLibrary** by calling a suitable method.
   c) borrow the first book from **myLibrary** by calling a suitable method.
   d) call the method **saveIntoFile()** to save the required information.

Question 6: [13 marks: 7 + 6]

Based on the classes of Part 3 - Question 5,

Given the following code which displays the following GUI application:




```
public void start(Stage primaryStage) {
 //Missing Code (1): Initialization

   Library myLibrary = ……

   Label isbnL = ……
   Label titleL = ……
   Label noOfPagesL = ……
   Label quantityL = ……

   TextField isbnTF = ……
   TextField titleTF = ……
   TextField noOfPagesTF = ……
   TextField quantityTF = ……

   Button addB = ……

   GridPane pane = ……
   primaryStage.setTitle(……);

   pane.addColumn(0, isbnL, ……, ……, ……, ……);
   pane.addColumn(1, isbnTF, ……, ……, ……);

   Scene scene = new Scene(……);
   primaryStage.setScene(……);

   primaryStage.show();

 //Missing Code (2): Clicking on the button
 ……
 ……
 ……
}
```

1. Complete the first missing code to initialize the UI controls.
2. Complete the second missing code such that when the user fills in the required data and clicks the button **"Add Book"**, a `Book` object will be created (using the first 3 fields) and added with its quantity (in the fourth field) to `myLibrary`.

End of Questions

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